NewsLens: Explainable News Classification

Problem: a topic label alone does not tell a user when to trust or review a result.

Pipeline: English article → three class scores → relative terms → accept or manual review.

Yi Tan (2312260). The live demo uses a local lexical model.

Background and Questions

Can a compact local model classify news and show its own evidence?

Does calibration improve probability quality without changing the predicted class?

What coverage does a review threshold retain, and how can explanation arithmetic be checked?

System and Evaluation Protocol

11,519 labeled articles; shared train / validation / test counts: 9,215 / 1,152 / 1,152.

Seed 42 and saved row identifiers; vocabulary, IDF, and NYT embeddings use training text only.

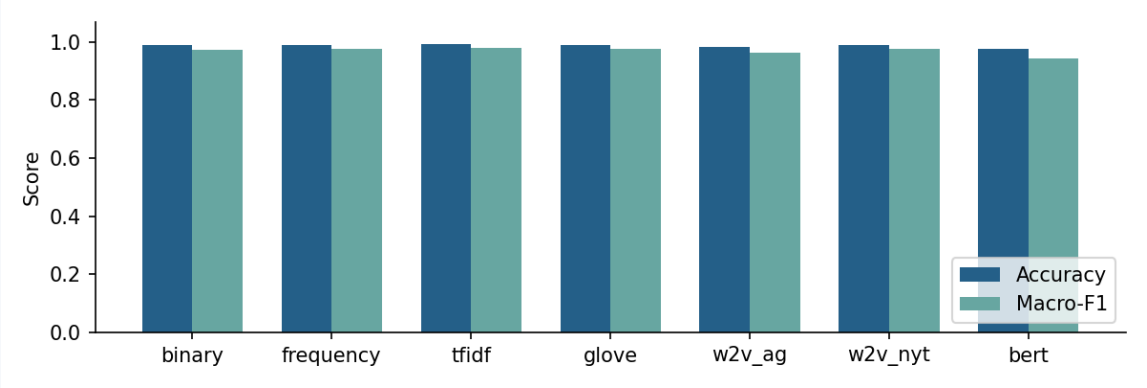
Original components: calibration, threshold selection, additive explanation, deletion check, interface, and tests.

Seven Representation Experiments

All completed scores are read from the same 1,152 test rows.

The TF-IDF baseline reaches 99.13% accuracy and 98.03% macro-F1.

Averaged embeddings and BERT differ in training corpora and visible input length.



Decision Policy

Use one half of validation to fit temperature and the other to choose a review threshold.

Keep test labels outside model, checkpoint, temperature, and threshold selection.

Term contribution is the feature value times the difference of two class coefficients.

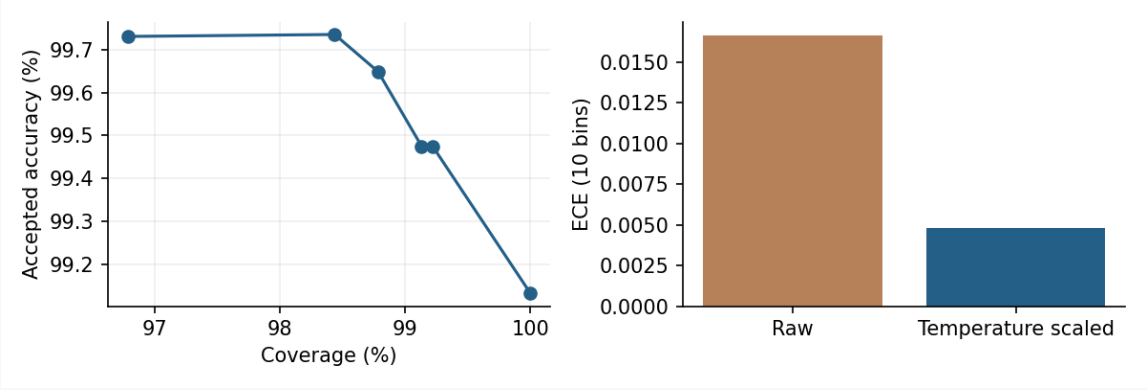
Calibration and Explanation Tests

Test ECE before / after scaling: 0.0167 / 0.0048.

Selected threshold: 98.44% coverage; 99.74% accuracy among accepted articles.

Top-five feature deletion changes 7.00% of predictions; random deletion changes 0.00%.

Contributions explain linear scores; they are not causal facts about the news.



Live Demonstration

Run Launch_Demo.ps1 and open the local browser page.

Analyze a sports story; compare three lexical models and inspect positive term contributions.

Try an empty or all-unknown input to observe clear validation or review behavior.

Limitations

Sports dominates the data; inspect macro-F1 and minority-class errors.

Removing verbatim train/test duplicates does not remove event-level near duplicates.

All test stories exceed 64 WordPiece positions, so BERT sees only a short prefix.

No cross-domain abstention guarantee or human explanation study is claimed.

Conclusion and Contribution

NewsLens provides a tested local classification, calibration, explanation, and review workflow.

Saved splits and per-example predictions make the scores independently checkable.

Yi Tan is the sole member; libraries, pretrained models, and Codex assistance are disclosed.